

| Assessment | Test | Domain        | Skill                  | Difficulty                                          |
|------------|------|---------------|------------------------|-----------------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | Medium <div><div></div><div></div><div></div></div> |

Question

Which expression is equivalent to  $(8x^3 + 8) - (x^3 - 2)$ ?

Answer

- A.  $8x^3 + 6$
- B.  $7x^3 + 10$
- C.  $8x^3 + 10$
- D.  $7x^3 + 6$

Correct Answer: B

Rationale

Choice B is correct. The given expression is equivalent to  $8x^3 + 8 - x^3 - (-2)$ , or  $8x^3 + 8 - x^3 + 2$ . Combining like terms in this expression yields  $7x^3 + 10$ .

Choice A is incorrect. This expression is equivalent to  $(8x^3 + 8) - 2$ , not  $(8x^3 + 8) - (x^3 - 2)$ .

Choice C is incorrect. This expression is equivalent to  $(8x^3 + 8) - (-2)$ , not  $(8x^3 + 8) - (x^3 - 2)$ .

Choice D is incorrect. This expression is equivalent to  $(8x^3 + 8) - (x^3 + 2)$ , not  $(8x^3 + 8) - (x^3 - 2)$ .